

1. Business Problem & Objectives

Product returns create direct refund exposure and can also create downstream operational costs. The objective is to quantify the return problem, understand the main return drivers, identify where risk is concentrated, and recommend targeted interventions.

Business Questions

- How should return reasons be categorized and counted?
- Which product categories have the highest return rate?
- Which return reason is most common?
- Are there meaningful price-range and monthly/time patterns?
- What are the likely root-cause areas?
- Which fixes should be implemented first and how should success be measured?

2. Data & Analytical Approach

Layer	What was done	Business purpose
Categorisation	Standardized return reasons and counted returned units	Identify biggest problems
Pattern analysis	Compared categories, products, price ranges and time	Locate concentration and risk
Prioritization	Used return volume, return rate and refund exposure	Focus effort on high-impact issues
Root-cause assessment	Converted patterns into hypotheses	Define validation needs
Recommendations	Assigned an action and KPI to each priority	Translate analysis into decisions

3. Data Quality & Assumptions

Item	Value
Raw rows	4,286 order-line rows
Unique orders	3,100
Returned orders	745
Return rate	24.03%
Returned units	1,113
Returned products	104
Returned-order logic	Return_Quantity > 0
Return Days	Return Date - Delivery Date
Refund figure used	■44.07L from Power BI / closure report

Note: an earlier analysis note contains a different derived refund total. This final business report uses ■44.07L consistently so the executive narrative has one source of truth.

4. Return Categorisation & Pattern Analysis

Returned-unit counts were used for prioritizing operational burden.

Return reason	Returned units	Share
Product Not As Expected	228	20.5%
Wrong Size	217	19.5%
Damaged Product	184	16.5%
Quality Issue	143	12.8%
Wrong Item	108	9.7%
Defective Product	86	7.7%
Late Delivery	76	6.8%
Changed Mind	71	6.4%

Category return-rate pattern

Category	Return rate
Fashion	26.1%
Electronics	21.7%
Home & Kitchen	16.6%
Sports	16.0%
Beauty	14.3%
Books	11.8%

Fashion has the highest category return rate at 26.1%. Electronics follows at 21.7%. Price-range and monthly views are used as diagnostic context and should not be treated as causal evidence.

5. Top 3 Problems & Root-Cause Hypotheses

Problem	Evidence	Likely area	Status
Product Not As Expected	228 units; ■7.95L	Listing / PDP content	Hypothesis
Wrong Size	217 units; 192 (88%) Fashion; ■5.87L	Listing / Product sizing	Hypothesis
Damaged Product	184 units; 139 (76%) Electronics + H&K; ■9.36L	Packaging / Logistics	Hypothesis

Root-cause guardrail: the dataset does not contain PDP-content versions, size-chart versions, carrier, packaging-type or damage-location fields. Therefore the analysis identifies likely investigation areas, not proven causes.

6. Recommendations & Success Measurement

Priority	Problem	Fix	Owner	Proposed target	Measurement
1	Product Not As Expected	Audit and rewrite descriptions/images for the top 10 SKUs. If on this reason for return, replace	Marketing	Proposed: -15% returned units	Compare fashion-specific returned units
2	Wrong Size	Rebuild Fashion size charts and add fit guidance. Proposed: reduce Fashion returns by 15%	Fashion Merchandising	Proposed: reduce Fashion returns by 15%	Compare from 26% to 11% of total returns
3	Damaged Product	Upgrade protective packaging and audit courier partners. Proposed: -20% Damaged Products	Operations / Electronics + Pkgs	Proposed: -20% Damaged Products	Compare fundamental damage rate

Business Impact

Measure	Value
Total refund exposure	■44.07L
Average monthly refund exposure	~■3.67L (annual total / 12)
Refunds as % of Net Sales	20.5%
Top 3 returned units	629 (56.5%)
Top 3 refund exposure	■23.18L (52.6%)

Important: proposed KPI targets are management targets, not historical results. No realized savings are claimed because no intervention was executed.

7. AI-Assisted Workflow

Stage	AI / MCP contribution	Analyst responsibility
Model inspection	Inspect tables, grain and relationships	Confirm scope and meaning
DAX support	Develop/debug measures	Define KPI logic and validate
Pattern analysis		